Jia-Le Mou 牟嘉乐

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Research Interests

I use geochemical analyses integrated with numerical and computational modeling, thermodynamics, and statistical approaches to address the following research questions:

- Mantle structure and thermal conditions: I explore the composition, thermal structure, and evolution of Earth's mantle, and how these relate to tectonics and crust formation, using geochemical datasets combined with parameterized modeling.
- Crustal differentiation and thermal evolution: I study how crustal processes influence the redistribution of radiogenic elements and then the thermal structure and stability of the continental crust, using numerical modeling and geochemical constraints.
- Application of two-phase flow in porous media to magma chamber processes: I apply numerical and computational tools to model melt migration and segregation in magma systems, with implications for ore formation.

Education

Rice University, PhD in Geology — 09/2022-expected 08/2026
Thesis: "Thermal controls on global tectonics and continental evolution from mantle melting records"
Advisor: Cin-Ty Lee

University of Science and Technology of China, B.S in Geochemistry — 09/2018-06/2022
Thesis: "The role of magmatic-hydrothermal processes on the genesis and evolution of granites"
Advisor: Fang Huang

Working Experience

Research Assistant, Rice University — 08/2022-present
Research Assistant at Key Laboratory of Crust-Mantle Materials and Environments, USTC — 06/2019-06/2022

Grants, Honors & Awards

Torkild Reiber Award in Geology — 05/2026
CPO2H Graduate Fellowship — 12/2024
Alison Henning Teaching Award — 01/2024
National Encouragement Scholarship — 11/2020
National Encouragement Scholarship — 11/2021
Endeavour Scholarship — 11/2020

Publications

1. **Mou, J.-L.**, Lee, C.-T., & Borchardt, J. (2025). Calibrating olivine forsterite content as a measure of melting degree in residual peridotites. Geochimica et Cosmochimica Acta. https://doi.org/10.1016/j.gca.2025.08.021
2. **Mou, J.-L.** & Lee, C.-T. A step change in Earth's thermal history driven by the onset of plate tectonics. Proceedings of the National Academy of Sciences. Revision submitted.
3. **Mou, J.-L.**, Lee, C.-T., & Zhang, J. The role of crustal thickness in governing heat-producing element distribution and cratonization. Manuscript completed, in internal review.
4. Lee, C.-T., Zhang, J., Keller, D., Zhang, Y., Ji, D., **Mou, J.-L**, (2026). Extreme magmatic differentiation and silicic magmatism without low melt fractions. Lithos 536–537, 108581. https://doi.org/10.1016/j.lithos.2026.108581

5. Lee, C.-T., Zhang, Y., Namur, O., **Mou, J.-L.**, Bakhshian, S., Keller, D., Sun, C. Layered mafic intrusions, magmatic turbidites, and the making of platinum ores. Proceedings of the National Academy of Sciences. Under review.

Conference Abstract

Jia-Le Mou, Cin-Ty Lee. A step change in Earth's thermal history driven by the onset of plate tectonics [Poster]. AGU Fall Meeting, 2025.
Jia-Le Mou, Cin-Ty Lee. A step change in Earth's thermal history driven by the onset of plate tectonics [Poster]. Gordon Research Conference, 2025.
Jia-Le Mou, Jackson Borchardt, Cin-Ty Lee. Calibrating olivine Forsterite content as a measure of melting degree in residual peridotites [Oral presentation]. Goldschmidt Conference, 2024.
Jia-Le Mou, Dingsheng Jiang, Gengxin Deng, Fang Huang. Ba isotope constraints on the genesis of the Koktotay pegmatite associated with giant metal deposits [Poster]. 18th Annual Academic Conference of the Chinese Society for Mineralogy and Geochemistry, 2021.

Field Work Experience

Southern Oregon, USA (1 week, 2024, collection of integrated geologic and geophysical data)
New Mexico, USA (3 days, 2023; igneous petrology, structure/tectonics, as a TA)
New Mexico, USA (1 week, 2023; sedimental petrology)
Utah State, USA (3 days, 2022, igneous petrology, structure/tectonics)
Altay Region, Xinjiang, China (2 weeks, 2021; sampling, igneous petrology, structure/tectonics)
Dabie Mountains, Anhui, China (1 week, 2021; metamorphic petrology, igneous petrology, tectonics)
Tan Lu Fault Zone, Anhui, China (1 week, 2021; structural geology, tectonics, metamorphic petrology)
Chaohu Lake, Anhui, China (1 week, 2019; sedimentary petrology, stratigraphy, structural geology)
Beidaihe, Hebei, China (1 week, 2019; igneous petrology, sedimentology)
Loess Plateau Region, Shaanxi; Laohugou Glacier No.12, Gansu, China (1 week, 2019; paleoclimate, climate)

Teaching

EEPS 101 The Earth — Spring 2026
Teaching assistant
EEPS 110 Earth, Environment and Society — Spring 2025
Teaching assistant
EEPS 322 Earth and Planetary Material — Fall 2023
Teaching assistant

Service & Outreach

EEPS Open House outreach event, EEPS, Rice University — Spring 2026
EEPS Open House outreach event, EEPS, Rice University — Spring 2025
Secretary of GeoUnion, EEPS, Rice University — 2023-2024
Captain of Women's Football Team of School of Earth and Space Sciences, USTC — 09/2020-12/2021

Skills

Analytical instruments: EPMA, SEM, MC-ICP-MS, ICP-MS
Programming & modeling: Python, OpenFOAM (C++)
Isotope geochemistry & sample preparation: column chemistry (Fe & Ba isotopes)